BE (E&TC)	PLC SCADA and Automation (Elective-III)	
Experiment No: 3	To implement and verify the output of a given Boolean expression using PLC programming and test it in both software simulation and hardware.	**Pages**

AIM

To implement and verify the output of a given Boolean expression using PLC programming and test it in both software simulation and hardware.

APPARATUS / SOFTWARE REQUIRED: TIA 13(Totally Integrated Automation)

Sr. No.	Equipment / Software	Specification / Example
1	PLC Trainer Kit	Siemens S7-1200 / Allen Bradley MicroLogix / Delta PLC
2	PLC Programming Software	TIA Portal / RSLogix / WPLSoft
3	Push Button Switches	NO / NC as per requirement
4	Indicator Lamps / LEDs	24V DC
5	Connecting Wires	As required
6	Power Supply	24V DC

OBJECTIVES

- To understand the conversion of Boolean expressions into PLC ladder logic.
- To implement logical operations using PLC programming instructions.
- To verify the correctness of the Boolean expression through simulation and hardware testing.

THEORY

A **Boolean expression** represents a logic statement using logical operators:

- **AND (·)** – Output is 1 only when all inputs are 1.
- **OR (+)** – Output is 1 when any input is 1.
- **NOT ($\bar{}$)** – Output is the complement of input.

PLC programming supports Boolean logic directly in **Ladder Logic**, **Function Block Diagram (FBD)**, or **Statement List (STL)**.

By translating the Boolean expression into a ladder logic program, we can simulate and test it on a PLC.

PROBLEM STATEMENT

Example Boolean Expression:

Y=(A·B)+(C‾·D)Y = (A \cdot B) + (\overline{C} \cdot D)Y=(A·B)+(C·D)

Where:

- A, B, C, D → Inputs (Push Buttons/Switches)
- Y → Output (Lamp/LED)

PLC Address Assignment

Variable	PLC Address	Description
A	I0.0	Input Switch A
B	I0.1	Input Switch B
C	I0.2	Input Switch C
D	I0.3	Input Switch D
Y	Q0.0	Output Lamp

Ladder Logic Implementation

Logic Steps:

1. Implement **A AND B** → First rung.
2. Implement **NOT C AND D** → Second rung.
3. Combine the two results using **OR**.

Ladder Diagram Example

Rung1: [ ]A[ ]B----( ) M0.0 (Intermediate Result1)
Rung2: [/]C[ ]D----( ) M0.1 (Intermediate Result2)
Rung3: [ ]M0.0OR[ ]M0.1----( ) Y

PROCEDURE

Software Implementation:

1. Open PLC programming software (e.g., TIA Portal).
2. Create a new project and configure PLC type.
3. Assign input and output addresses as per design.
4. Implement ladder logic according to the Boolean expression.
5. Compile and download the program to the PLC or simulator.
6. Test the logic by changing input values.

Ladder diagram:

Hardware Implementation:

1. Connect push button switches to PLC input terminals.
2. Connect the output lamp to PLC output terminal.
3. Power on the PLC and test by pressing the switches according to the truth table.

Truth Table for Example Expression

A	B	C	D	Output Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	1	1
1	0	0	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

OBSERVATIONS

- Verify each row of the truth table through simulation and hardware.
- Check if the output matches expected results.

Result

The given Boolean expression was successfully implemented in PLC using ladder logic and verified through both simulation and hardware.

Applications

- Digital control systems.
- Industrial automation logic design.
- Machine safety circuits.

CONCLUSION

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Questions:

1. What is the difference between SOP (Sum of Products) and POS (Product of Sums) forms?

2. Why do we implement Boolean expressions in PLC instead of hardwired circuits?

3. Draw the truth table for the given Boolean expression $Y=A\cdot\bar{B}+CY = CY=A\cdot B+C$.

REFERENCES:

[1]Petruzella, F. D. (2017). *Programmable Logic Controllers* (5th ed.). McGraw-Hill Education.
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[3] Siemens AG. (2023). *TIA Portal Programming Manual*.Siemens Industry Online Support